Anatomy of the Treasury Market

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Abstract

The Treasury market's central role in the financial system rests on its liquidity and its safe-haven property. Factor regressions can track such properties but cannot say which investors drive them. We develop an equilibrium framework that decomposes yield-factor regressions into investor-level drivers, and an optimal estimator that recovers each investor's responses to yields and factors. Using sector-level flow data spanning 1986–2023, we show that both properties are eroding as private investors such as dealers and foreigners step back, and that the Federal Reserve has partially stepped in their place, moderating the aggregate decline.

Keywords: Treasury market; demand system; price elasticity; liquidity; safe haven; flight to safety; granular instrumental variables.

JEL codes: E43, E58, G11, G12, H63.

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1 Introduction

The U.S. Treasury market is the bedrock of the global financial system. Two properties of this asset class help underpin that position. First, the market is deep and liquid: it can accommodate large shifts in investor demand without large price impact, which makes Treasuries the natural home for funds that investors worldwide may need to deploy quickly. Second, it is a safe haven: it provides insurance by appreciating in periods of economic stress. There is growing concern that both properties are eroding. Foreign and domestic private investors are stepping away from the market, and in March 2020, at the height of the COVID crisis, Treasuries fell in value just when a safe haven should rise. If these properties go, the central role of the Treasury market may come into question. Both properties are equilibrium outcomes: they reflect how the investors who hold Treasuries respond to prices and to risk. Whether these properties are eroding therefore depends on which investors sustain them, and on whether their behavior is changing.

Identifying those investors requires a framework that links the market's aggregate properties to the investors behind them. We develop one. It connects investors' portfolio choices to the aggregate pricing properties that standard factor regressions measure. A factor regression records how much Treasury prices move with a factor—for instance, how much they rise when market stress increases. Its loadings, however, blend two things: how much investors' demand shifts when the factor moves, and how elastically the market absorbs those shifts, which is its liquidity. The regression cannot tell the two apart, let alone say which investors are behind each. Our framework separates them and attributes each to the investor groups behind it. To estimate the model, we develop a flexible optimal estimator that needs only sector-level holdings and flows.

We apply the framework to four decades of data and find that both the liquidity and the safe-haven property are eroding. The erosion we observe is the net of two moves. A handful of private investor groups have stepped back, and the Federal Reserve (Fed) has partially stepped in. First, the market's liquidity has declined moderately. Private investors, foreign investors in particular, have pulled back considerably from providing liquidity, yet aggregate liquidity has fallen far less, because the Federal Reserve has filled much of the gap they left. Second, the safe-haven property has visibly faded. Treasuries became a safe haven only in the late 1990s and have since lost much of that property; dealers, not foreign investors, drove both the rise and the decline, and the Federal Reserve's response to stress slowed the fall without stopping it.

Our framework starts from the portfolio choice of each investor. We model each

sector's demand for Treasuries as a log-linear function of prices, economic factors, and sector-specific shocks. This form is a first-order approximation to a broad class of portfolio choice problems, so differences in investors' objectives, constraints, and beliefs show up as differences in their estimated price elasticities and factor loadings, without our having to model those differences explicitly. Because our question concerns the liquidity and safety of the Treasury market as a whole, we treat Treasuries as a single asset class and abstract from substitution along the yield curve. Aggregating these investor demands through market clearing gives us the standard price-factor regression, and shows what its loadings measure. When a factor moves, investors' demand shifts, and prices must move until the market absorbs the net shift. An aggregate price loading therefore combines the direct shift in demand with the price move needed to absorb it, and that price move is governed by the market's aggregate price elasticity, which is its liquidity.

Because we estimate demand investor by investor and link it to this aggregate pricing equation, we can decompose both the direct factor driven shift and subsequent liquidity provision into investor contributions. The aggregate direct response to an innovation in a factor is the size-weighted sum of investors' individual responses. This decomposition reveals which investors drive that response. For example, it shows which investors increase or reduce their Treasury demand during episodes of financial tightening. The aggregate price elasticity is likewise the size-weighted sum of investors' elasticities, so a large sector that does not respond to price adds little liquidity. Both are investor-level objects that we allow to change across regimes, so we can trace shifts in the aggregate price loadings, such as the rise and decline of the safe-haven property, to the changes in investor-level behavior that produced them.

Estimating these investor-level responses raises the central identification challenge. When a factor moves, investor holdings respond through two channels: directly, because the factor shifts their demand, and indirectly, because other investors' demand shifts move prices until the market clears. An investor may therefore sell Treasuries during a flight to safety not because its own response to stress is negative, but because others' purchases have raised prices. Distinguishing the two channels requires an instrument for the price in every sector's demand equation. Existing Treasury market instruments, such as regulatory changes and monetary policy surprises, cover particular sectors or short episodes. We instead develop an optimal generalized method of moments (GMM) estimator that uses the idiosyncratic component of each sector's demand as an instrument in the other sectors' equations. The identifying assumption is that these components are uncorrelated across sectors once common factors are accounted for. The estimator is an

optimal implementation of granular instrumental variables (GIV; Gabaix & Koijen, 2024). It has two advantages. First, it lets each sector have its own price elasticity and factor loadings. Second, by giving more weight to demand shocks from the sectors with greater influence on prices, it delivers much more powerful instruments. That statistical power makes it possible to estimate how price elasticities and factor loadings differ across sectors, and how they change over time.[1]

Our data are publicly available sector-level holdings of Treasuries, drawn from the Financial Accounts, Treasury International Capital data, and Call Reports, covering 1986 to 2023. The market's liquidity and its safe-haven property both change slowly, over years rather than quarters, and this long sample lets us follow them, and the investors behind them, through several distinct regimes. We take the two properties in turn.

The market's liquidity has declined moderately. Before the financial crisis, a sale of Treasuries equal to 1% of the market lowered their prices by about 0.9%, or equivalently raised their yields by about 16 basis points at an average duration of around 5.5 years; since then, it has lowered prices by about 1.2%, or 22 basis points in yields. This is a moderate decline, given how much the investor base has changed since the financial crisis.

Private investors have retreated from providing liquidity, and the Federal Reserve has filled much of the gap. Domestic institutions such as banks and pension funds held much of the market in the 1990s; by 2008, foreign investors held nearly two-thirds of it. After the financial crisis, however, the size of foreign holdings no longer translated into liquidity provision. Foreign holdings have declined steadily since 2008 but remain sizeable, at about a third of the market in 2023. Their price elasticity fell from 0.52 to 0.09, so these holdings now absorb little of the market's demand shifts. China, once the largest foreign holder, has cut its Treasury holdings by about 40% since 2013. The Federal Reserve filled this gap, moderating the aggregate decline in liquidity: its post-crisis holdings now account for about a third of the market's aggregate elasticity. Without them, and holding other investors' behavior fixed, the market's liquidity would have eroded far more: the same 1% sale would today lower prices by about 1.8% (33 basis points in yields), twice its pre-crisis impact. As a result, the Treasury market's liquidity now rests partly on the central bank rather than on private investors alone.

The safe-haven property emerged in the late 1990s and has faded since the early 2010s. Before 1997, Treasury prices did not rise when market stress increased, as mea-

[1]We make the estimator available as open-source packages for Julia (https://github.com/FuZhiyu/OptimalGIV.jl) and Python (https://github.com/FuZhiyu/optimalgiv).

sured by jumps in stock market volatility. From 1997 through the financial crisis, they did: a one-standard-deviation rise in volatility raised Treasury prices by about 1.5% within the quarter, pushing yields down by some 27 basis points. Since 2012, this response has largely faded, to less than a third of its peak by the end of our sample. The rise and fall echoes the well-documented shift in the stock–bond correlation and the concerns raised during the COVID-19 crisis about Treasuries' waning safety (Campbell, Pflueger, & Viceira, 2020; Antolín-Díaz, 2026; He, Nagel, & Song, 2022; Vissing-Jorgensen, 2021).

Foreign investors did not drive Treasuries' safe-haven property. A common view attributes the appreciation of Treasuries during periods of stress to foreign investors' flight to safety (Jiang, Krishnamurthy, & Lustig, 2021, 2024). Our demand system separates investors' demand responses to uncertainty from their responses to price changes. We find that foreign investors contributed little when the safe-haven property emerged in the late 1990s, and that since the financial crisis their demand has, if anything, declined with uncertainty. These findings support a precautionary saving view in which foreign investors accumulate Treasuries in normal times and draw on them for liquidity during periods of stress, benefiting from their insurance value without generating the countercyclical demand that sustains it (Eisenbach & Phelan, 2026; Bianchi, Hatchondo, & Martinez, 2018; Devereux, Engel, & Wu, 2023). Reduced-form evidence points in the same direction: after adjusting for growth in Treasury supply, foreign flows decline with uncertainty while domestic private flows rise, including when the sample is restricted to the global financial crisis.

Dealers drive both the rise and the decline of the safe-haven property. Before the late 1990s, dealers' demand for Treasuries showed little response to market stress. From then until the financial crisis, dealers bought Treasuries when stress rose, and afterwards this response fell back to about zero. After dealers stepped back, the safe-haven property survived for several more years, carried by the household sector—a group that includes hedge funds—and by the Federal Reserve's asset purchases in response to stress. After the household sector's responsiveness declined towards the end of our sample, the safe-haven property is almost entirely driven by the Fed's policy support.

Together, these findings show that the investors sustaining the Treasury market's central properties have changed, and that the Federal Reserve now stands behind both. It has offset much of the private retreat from liquidity provision, and its response to stress helped carry the safe-haven property after dealers withdrew. This poses a challenge for the design of monetary policy normalization strategies. If the Federal Reserve shrinks its balance sheet and private investors do not return, the market may see a further, and

larger, decline in its ability to accommodate investor demand shifts, and its safe-haven property may not return.

1.1 Related Literature

A burgeoning literature studies how investors' demand shapes the pricing of U.S. Treasuries. Our contribution is to decompose the market's two defining properties, its liquidity and its safe-haven role, into the contributions of investors and to follow those contributions over four decades.[2] Koijen, Koulischer, Nguyen, and Yogo (2017) employ a demand system for government bonds in the euro area to examine the portfolio rebalancing mechanism of quantitative easing; Jiang, Richmond, and Zhang (2024) and Jiang, Richmond, and Zhang (2025) combine cross-country bilateral portfolio holdings data with a global demand system to understand the exorbitant privilege the U.S. commands on its debt and the strength of the U.S. dollar. Focusing on the U.S. Treasury market, Cavaleri (2023) studies different sectors' demand for convenience services, Eren, Schrimpf, and Xia (2026) estimate sectoral elasticities for Treasuries using monetary policy surprises, and Jansen, Li, and Schmid (2024) study how sector-specific demand shocks propagate through the term structure via risk-averse arbitrageurs and habitat investors who substitute across maturities. We contribute to this literature on two fronts: we trace the market's liquidity to the investors that provide it and show how their roles have shifted over four decades, and we identify the investors that make Treasuries a safe haven.

Among demand-based studies of the Treasury market, our work is closest to Jansen et al. (2024) (henceforth JLS). The key difference is the primary focus. Using maturity-bucket holdings from 2011, JLS study substitution along the Treasury yield curve and the effects of monetary policy on term premia. We instead document the retreat of private investors from liquidity provision and the rise and fall of the safe-haven property.[3] These properties shift across regimes that span decades, so we follow the market back to the 1980s, and our optimal GIV estimator provides the power to estimate how investors' behavior changes between regimes. The trade-off is that the longer sample relies on coarser

[2]Many recent advances in this literature build on the demand-system approach to asset pricing first applied to the cross-section of stocks (Koijen & Yogo, 2019). Variants have since been applied to the equity market (Gabaix & Koijen, 2022), the corporate bond market (Chaudhary, Fu, & Li, 2026a; Bretscher, Schmid, Sen, & Sharma, 2026; Fang & Xiao, 2025), and exchange rates (Camanho, Hau, & Rey, 2022; Koijen & Yogo, 2026; An & Huber, 2024). We contribute to this broader literature by proposing a novel identification method, described below.

[3]Eren et al. (2026) also estimate sectoral demand elasticities, for several countries, using monetary policy surprises as instruments. Their application focuses on the interaction between investor composition and central bank asset purchases and sales rather than on the evolution of the market's liquidity and safe-haven property.

data, so we treat Treasury notes and bonds as a single asset class. The different focuses drive different methodological choices. To study how demand shocks propagate across the yield curve, JLS take a structural approach, placing arbitrageurs (dealers and hedge funds) at the center of their model alongside estimated demand curves for the other sectors. To decompose aggregate price loadings on factors into sector contributions, we take a more agnostic approach: we estimate every sector's demand with the same flexible specification, dealers included, and let price elasticities and factor loadings change across regimes. This flexibility matters for our results, since dealers turn out to be the largest contributor to both turns in the safe-haven property. The two papers thus answer complementary questions: JLS quantify how sector-specific demand shocks propagate across maturities and into term premia, while we document which investors have supplied the market's liquidity and its safe-haven property, and when they withdrew.

Our first contribution concerns the market's liquidity. Our aggregate estimate is in line with the literature that measures the price impact of demand or supply shifts through event studies, term-structure models, and auction bidding data (e.g., Bernanke, Reinhart, & Sack, 2004; Warnock & Warnock, 2009; Greenwood & Vayanos, 2010; Gagnon, Raskin, Remache, & Sack, 2011a; Krishnamurthy & Vissing-Jorgensen, 2012; Hanson & Stein, 2015; Kaminska & Zinna, 2020; Jansen et al., 2024; Somogyi, Wallen, & Xu, 2025); Appendix B.3.1 maps these estimates into macro multipliers and compares them with ours. We depart from this literature in two ways. First, we decompose the aggregate elasticity into the contributions of individual sectors, whereas existing work studies one investor group at a time, such as the Federal Reserve (Haddad, Moreira, & Muir, 2024; Krishnamurthy & Vissing-Jorgensen, 2011), banks (Jiang, Matvos, Piskorski, & Seru, 2024), mutual funds (Selgrad, 2023; Huang, Jiang, Liu, & Liu, 2025), insurance companies (Chaudhary, 2024; Koijen & Yogo, 2023), broker-dealers (Du, Hébert, & Li, 2023; Vayanos & Vila, 2021; Greenwood & Vayanos, 2014), and foreign investors (Beltran, Kretchmer, Marquez, & Thomas, 2013; Wolcott, 2020; Ahmed & Rebucci, 2024; Tabova & Warnock, 2026). Our equilibrium model estimates all sectors jointly and separates shifts in investors' demand from their responses to equilibrium price changes, which reduced-form analyses of observed holdings cannot disentangle. Second, because each sector's elasticity can change across regimes, we can date the decline in aggregate liquidity and attribute it. Somogyi et al. (2025) document that auction demand has become markedly less elastic since 2010 and point to the retreat of foreign investors as one possible reason. Our estimates show the same decline for the market as a whole and identify the sectors behind it. Foreign investors' elasticity fell far more than their holdings shrank, no private sector has taken their place, and the Federal Reserve's balance sheet now accounts for a substantial share

of the market's capacity to absorb demand shifts, which speaks to the literature on quantitative easing and tightening (Gagnon, Raskin, Remache, & Sack, 2011b; Krishnamurthy & Vissing-Jorgensen, 2011; Du, Forbes, & Luzzetti, 2026).

Our second contribution concerns the safe-haven property. A common view attributes the appreciation of Treasuries in periods of stress to foreign investors' flight to safety (Jiang et al., 2021, 2024). Because our framework decomposes the market's response to uncertainty into investors' direct demand responses, net of their responses to the price changes that others' purchases induce, we can test this view directly. We find little support for it. Foreign investors contributed little when the property emerged in the late 1990s, and since the financial crisis their demand for Treasuries has, if anything, declined as uncertainty rises. This cyclicality is consistent with a precautionary saving view of foreign holdings, in which foreign investors accumulate Treasuries in normal times and sell them to raise liquidity during risk-off episodes (Eisenbach & Phelan, 2026; Bianchi et al., 2018; Devereux et al., 2023, among others). Our estimates instead attribute the safe-haven property primarily to domestic investors, and above all to dealers, whose response to stress switched on in the late 1990s and off after the financial crisis. This links the safety of Treasuries to the literature on intermediaries' balance-sheet capacity and its role in Treasury market functioning (He et al., 2022; Du et al., 2023; Duffie et al., 2023), and to accounts in which the safety premium on Treasuries is time-varying and tied to their value in bad states (Acharya & Laarits, 2026; Brunnermeier, Merkel, & Sannikov, 2024). These findings have implications for the sources and sustainability of the special status of U.S. safe assets (Gourinchas & Rey, 2022; Du, Keerati, & Schreger, 2025).

Methodologically, our approach extends the GIV literature (Gabaix & Koijen, 2024). GIV has been increasingly applied in macro-finance settings where market-level instruments are difficult to find (Kundu & Vats, 2026; Kubitza, Sigaux, & Vandeweyer, 2024; Chodorow-Reich, Gabaix, Koijen, & Viviano, 2024; Adrian, Grinberg, Liang, Malik, & Yu, 2022; Camanho et al., 2022). Our extension accommodates heterogeneous price elasticities and factor loadings across investors without parameterizing them on observable characteristics, which is what allows the sector-level and regime-level variation that the two preceding paragraphs rely on. We also propose an economically motivated optimal GMM estimator that assigns greater weight to shocks from investors with greater influence on equilibrium yields. This ensures instrument relevance—a key challenge in a market as large as U.S. Treasuries—and yields an efficient estimator. It builds on other efficient GIV variants, such as Qian (2024) and Baumeister and Hamilton (2023), which characterize optimality statistically rather than economically.

1.2 Roadmap

The remainder of the paper is organized as follows. Section 2 describes our data and introduces an equilibrium model of the Treasury market that can be taken to the data. Section 3 outlines our estimation and identification approach. Section 4 documents the trends in the market's liquidity provision, and Section 5 the trends in its safe-haven property. Section 6 concludes. The Online Appendix develops the general estimation framework and its properties (Appendix A) and provides the data details, robustness checks, and additional results (Appendix B).[4]

2 The Treasury Market: Data and Model Framework

Our analysis focuses on the market portfolio of Treasury notes and bonds (Treasury securities with maturities exceeding one year, referred to as *Treasury bonds*). Yields on this portfolio are important indicators for the overall funding costs of the U.S. government and serve as benchmarks for analyzing other long-term interest rates in the economy. For this purpose, we combine sectoral holdings data, to be introduced in Section 2.1, and an empirically tractable model framework that we outline in Section 2.2. Because our question concerns the liquidity and safe-haven property of U.S. government debt as a whole, we treat Treasury bonds as a single asset class and abstract away from modelling the entire yield curve. This lets us characterize how aggregate Treasury bond prices are shaped by investors' allocation and government issuance decisions. Our empirical framework, however, can accommodate multiple assets and maturities, and we formally discuss how accounting for substitutes affects the interpretation of our estimates in Section 2.2 and Appendix A.5.

2.1 Data

The primary source of our Treasury bonds holdings and issuance data is the Federal Reserve's Financial Accounts (Z.1). This dataset reports positions and transactions (flows) in Treasury notes and bonds at the investor sector level. We complement these data with more granular information for banks and the foreign sector. For banks, we use Call Reports to collect individual commercial banks' holdings of and flows in and out of U.S. Treasury securities, following the procedure of Jansen et al. (2024) to impute Treasury

[4]Additional results and discussions are available in the Supplementary Material at https://anatomyoftreasury.github.io/AnatomyOfTreasury/Treasury_GIV_supplement.pdf.

holdings from reported Treasury and agency securities. For the foreign sector, we draw on Treasury International Capital (TIC) data to disaggregate holdings by country.[5]

Figure 1 shows the evolution of the market value of Treasury bonds holdings from 1986 to 2023, our sample period. Treasury bonds are held by five broad groups: (i) banks; (ii) asset managers, including mutual funds, insurers, and pension funds; (iii) households and other U.S. investors, such as non-financial institutions and state and local governments; (iv) the Federal Reserve; and (v) foreign investors. Over our sample period, the total supply of Treasury bonds increased from about $1.2 trillion to more than $20 trillion, rising from roughly 25% of GDP to over 70%.[6] Over the same period, investor composition has shifted substantially. In the first half of the sample period, domestic institutional investors such as banks and pension funds account for significant market shares, while holdings have shifted towards foreign investors and the Federal Reserve in the latter half. The foreign share peaked in 2008 and has declined since.

We measure the Treasury bond yield using the CRSP US Treasury Database. Using security-level data, we calculate the average yield of a market-weighted portfolio of Treasury notes and bonds that have maturities exceeding one year. Average yields during our sample period fluctuate between 0.5 and 9.3 percent, while the average duration of the market portfolio ranges from 4.6 to 7.4 years.

2.2 An Equilibrium Model

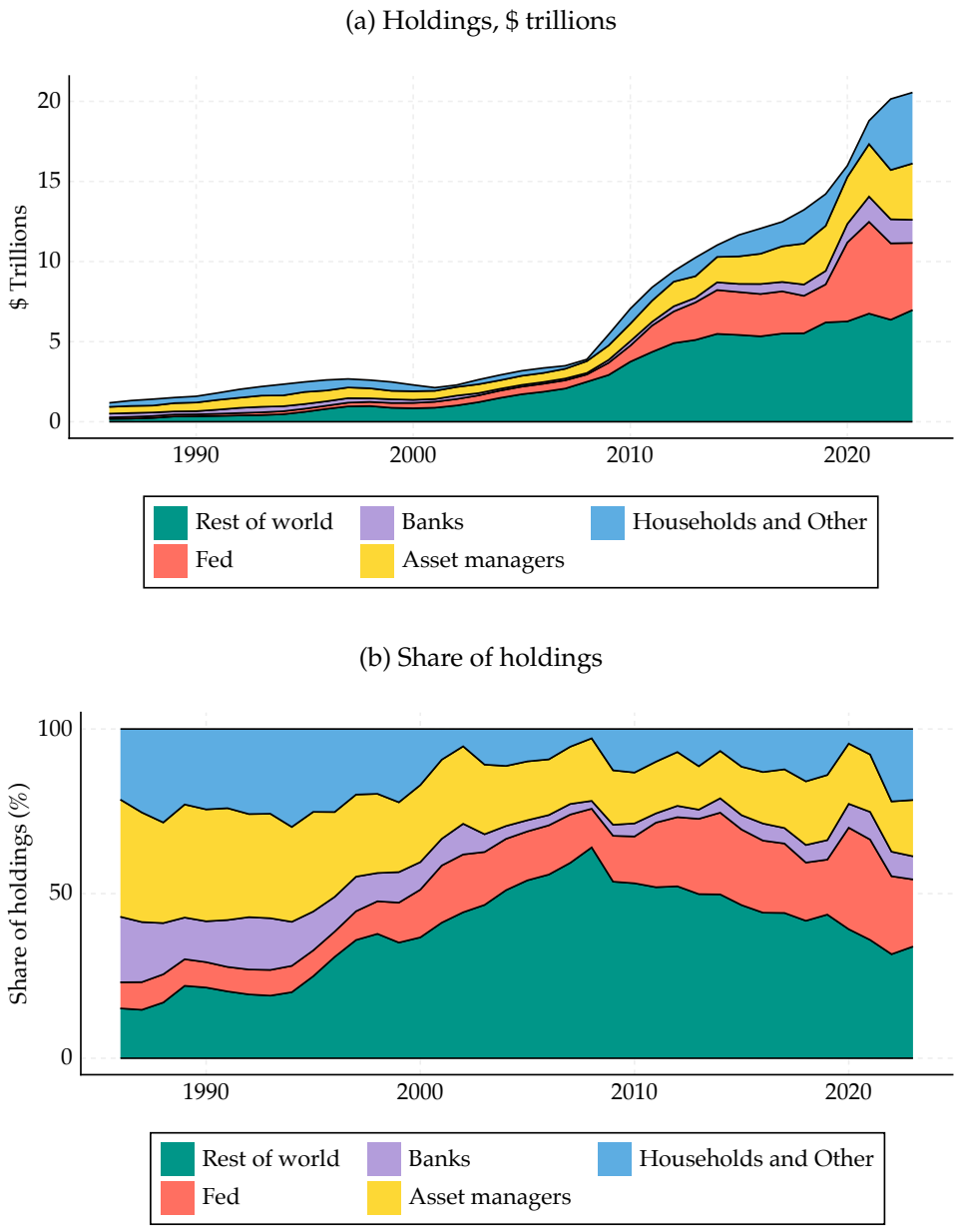
We outline our equilibrium model of the Treasury market. The model allows us to connect the yield we observe on Treasury bonds to the data we have collected on sectors' allocation decisions and the government's issuance decisions. The framework is parsimonious yet sufficiently flexible to incorporate sector-specific differences in objectives, constraints, beliefs, and substitution patterns across assets. Its flexibility also allows us to track the evolution of investors' liquidity provision and demand loadings on macro-financial factors over time to highlight the implications of structural shifts in the Treasury market.

Notation. We index sectors by i (e.g., insurance companies, the Federal Reserve). Treasury supply is treated as one of the sectors with negative demand. For banks and foreign

[5]Appendix B.1 provides more details on data construction, including the use of the entire Treasury securities holdings for sectors that do not report breakdowns, the imputation procedure for individual bank holdings, and the adjustment of reported discrepancies.

[6]Total federal debt stood at about 120% of GDP at the end of 2023, including notes and bonds, bills, and non-marketable securities.

Figure 1: **Holdings by Group: U.S. Treasury Notes & Bonds**



Note: This figure plots the year-end market value of U.S. Treasury notes and bonds holdings by investor groups in dollars (panel (a)) and as shares of the total (panel (b)). We aggregate investor sectors reported in Financial Accounts (Z.1) to five investor groups. *Asset managers* are insurers, pension funds, mutual funds, exchange-traded funds (ETFs), and dealers. *Households and Other* are the Households and Other categories. Appendix Table B.1 maps sectors to categories.

investors, we disaggregate further to the individual bank or country level, and refer to these banks and countries also as sectors. We use bold symbols for vectors and matrices ($\mathbf{q}_t$) and regular symbols for scalars ($q_{i,t}$). Key variables include: $Q_{i,t}$ for sector i's Trea-

sury holdings in face value terms, $A_{i,t}$ for the holdings in market value terms, P_t for the Treasury market portfolio price, and y_t for the average yield.

Portfolio choice. Investor sectors allocate to Treasuries under distinct frictions. Hedge funds target short-horizon returns subject to Value-at-Risk (VaR) limits and redemption risk; insurers buy-and-hold to match long-duration liabilities under capital rules; banks hold Treasuries for liquidity management, compliance, and duration hedging. Fully microfounding each sector is unwieldy given these interacting constraints. In order to flexibly capture these diverse drivers of Treasury demand, we leverage the fact that the first-order approximation of a large class of portfolio choice problems can be expressed as a log-linear function of prices and demand shifters:

$$\Delta q_{i,t} = \overline{\Delta q}_i - \zeta_{i,t} \Delta p_t + \nu_{i,t}. \tag{2.1}$$

Here, $\Delta q_{i,t}$ is the net purchase of Treasuries of sector i, normalized by a size weight. Δp_t is the percent change in the price of the Treasury market portfolio, $\zeta_{i,t}$ is sector i's price elasticity, $\overline{\Delta q}_i$ is a sector-specific intercept, and $\nu_{i,t}$ is a demand shifter. This expression follows from log-linearizing a Treasury demand or supply function around the long-run mean and differencing. The approach lets the data capture the motives behind portfolio choice via elasticities and shifters, and it locally nests a wide range of models. In Appendix A.4, we discuss the structural interpretation of elasticity $\zeta_{i,t}$ and shifters $\nu_{i,t}$ in various portfolio choice models, ranging from standard mean-variance portfolio choice to demand for convenience and learning-from-price models.

Interpreting the elasticity ζ_i in a multi-asset setting. Equation (2.1) represents a single-asset demand curve. In Appendix A.5, we show that this formulation nests a richer multi-asset demand system in which Treasury holdings depend not only on their own price but also on the prices of substitute assets. The key insight is that, by imposing market clearing conditions for these substitutes, we can express their equilibrium prices as functions of Treasury price changes and demand shifts. This collapses the multi-asset system to a reduced-form single-asset demand curve with a *total elasticity* $\zeta_{i,t}$, which embeds the feedback effects from cross-asset substitution.

To gain intuition, consider substitution between cash Treasuries and Treasury futures as in basis trades. In this case, $\zeta_{i,t}$ captures the overall responsiveness of sector i's *cash Treasury* holdings to a 1 percent change in Treasury prices induced by Treasury demand shifts, accounting for the resulting equilibrium adjustments in futures prices. The

reciprocal of the aggregate elasticity, $\zeta_{S,t}^{-1}$, corresponds to the overall price impact of a unit demand shock to cash Treasuries. This contrasts with a partial own-price elasticity, which holds Treasury futures prices fixed. For many policy applications—such as quantitative easing or large-scale foreign official flows—the total elasticity is the relevant parameter, as policymakers care about overall Treasury price movements. Our approach is designed to answer such questions.[7]

The single-asset representation therefore aggregates the matrix of partial own- and cross-price elasticities into a single parameter $\zeta_{i,t}$, which reduces dimensionality and improves tractability.

Demand shifters $\nu_{i,t}$. The demand shifters $\nu_{i,t}$ have two components: common factors $\boldsymbol{\eta}_t$ that affect all sectors and idiosyncratic shocks $u_{i,t}$ specific to each sector. This gives us:

$$\Delta q_{i,t} = \overline{\Delta q_i} - \zeta_{i,t}\Delta p_t + \underbrace{\boldsymbol{\lambda}_{i,t}\boldsymbol{\eta}_t + u_{i,t}}_{\nu_{i,t}}. \tag{2.2}$$

Common factors $\boldsymbol{\eta}_t$ capture economy-wide drivers such as monetary policy changes, risk sentiment shifts, or drifts in the inflation expectation. While all sectors are exposed to these factors, they respond with different intensities captured by sector-specific loadings $\boldsymbol{\lambda}_{i,t}$. For example, insurance companies may be more sensitive to long-term interest rate changes than hedge funds. The idiosyncratic component $u_{i,t}$ captures sector-specific drivers such as regulatory changes, portfolio rebalancing needs, or liquidity constraints that are orthogonal to common factors.

Equilibrium. We connect sector-level demand and supply to Treasury yields through market clearing. In equilibrium, all Treasury bonds must be held by someone—the net flow across all sectors, including Treasury supply, must sum to zero. Each sector's flow $\Delta q_{i,t}$ is weighted by a size weight $S_{i,t}$ that is allowed to be time-varying to arrive at the market clearing condition:

$$\sum_i S_{i,t}\Delta q_{i,t} = 0.$$

[7] Our estimates do not speak to questions regarding substitutability between Treasuries and futures and the arbitrage frictions between these assets. These questions require explicitly modeling both assets.

Treasury price adjusts to clear the market. To find the equilibrium price, we substitute our demand equations into the market clearing condition. This gives us the pricing equation:

$$\Delta p_t = \frac{1}{\zeta_{S,t}} \left(\overline{\Delta q}_{S,t} + \underbrace{\boldsymbol{\lambda}_{S,t} \boldsymbol{\eta_t} + u_{S,t}}_{\nu_{S,t}} \right), \tag{2.3}$$

where the S subscript denotes size-weighted aggregation across all sectors. For example, $\zeta_{S,t} \equiv \sum_i S_{i,t} \zeta_{i,t}$ is the aggregate elasticity and $\boldsymbol{\lambda}_{S,t} \equiv \sum_i S_{i,t} \boldsymbol{\lambda}_{i,t}$ is the vector of aggregate factor loadings.

Equation (2.3) shows that Treasury price changes are determined by aggregate demand shifts, scaled by the inverse of the aggregate elasticity, $\zeta_{S,t}^{-1}$. We refer to this inverse elasticity as the *macro multiplier*—it captures the percentage change in Treasury prices from a 1% change in aggregate net demand. Importantly, this multiplier reflects the total equilibrium effect, including amplification from cross-asset substitution as Treasury price movements propagate through other markets and feed back into Treasury demand. We provide formal derivations based on a multi-asset demand system in Appendix A.5.

Once we obtain estimates for sector-specific elasticities $\zeta_{i,t}$ and factor loadings $\boldsymbol{\lambda}_{i,t}$, we can decompose any Treasury price movement into contributions from individual sectors and common factors. The contribution of sector i to the price change is given by:

$$\Delta p_{i,t} = \frac{1}{\zeta_{S,t}} S_{i,t} \left(\overline{\Delta q}_i + \boldsymbol{\lambda}_{i,t} \boldsymbol{\eta}_t + u_{i,t} \right). \tag{2.4}$$

This decomposition reveals which investors drive Treasury price movements during different episodes. For example, consider a positive inflation surprise that causes Treasury prices to decline. Our framework can determine both the direct effect of the inflation shock and which specific investors (banks, foreign central banks, etc.) amplify or dampen this shock through their trading responses.

To implement this decomposition, we need to estimate the potentially time-varying elasticities $\zeta_{i,t}$ and factor loadings $\boldsymbol{\lambda}_{i,t}$. This presents a challenge: prices and quantities are jointly determined in equilibrium, creating simultaneity bias in standard estimation approaches. We lay out our identification strategy in the next section.

3 Identification and Estimation

3.1 Specification

We estimate (2.2) on nearly four decades of data. Rather than hold the demand parameters fixed over the whole sample, we allow them to shift across regimes. Letting $r(t)$ denote the regime that quarter t belongs to, we estimate

$$\Delta q_{i,t} = \overline{\Delta q}_{i,r(t)} - \zeta_{i,r(t)} \Delta p_t + \boldsymbol{\lambda}_{i,r(t)} \boldsymbol{\eta}_t + u_{i,t}, \tag{3.1}$$

so that each sector's intercept, price elasticity and factor loadings can differ across regimes. Our baseline lets every parameter differ before and after the financial crisis; Section 4 reports these estimates.[8]

Sector definitions and groupings. Our sectoral definitions follow the Financial Accounts. Twenty-seven sectors hold Treasury securities during our sample period. We aggregate these sectors to 11 categories for interpretability, grouping similar investor types. For example, holdings of government pension funds and private pension funds are aggregated to a single "pension fund" category. The complete list of sectors and their mapping to the aggregated categories is provided in Appendix Table B.1. One sector worth highlighting in the Financial Accounts is the "household sector," which serves as the residual sector to ensure the sum of all sectors' flows equals zero. This sector should be broadly interpreted as a catch-all category for investors not included or observed in other sectors, such as hedge funds, endowment funds, and family offices.[9] In addition, we separately consider the Federal Reserve and the Treasury in this analysis, by estimating their price elasticities and factor loadings separately, in order to highlight the potential distinct roles played by each policy entity as buyers and sellers of Treasuries in shaping price dynamics.

Flows and positions. The flow measure $\Delta q_{i,t}$ is defined as the net dollar flows $F_{i,t}$ by investor i, normalized by the sector's size weight $S_{i,t}$ times the total market values of

[8] Before 2009, the Fed's outright purchases of Treasury bonds were small in scale relative to quantitative easing (QE), mostly associated with reserve-management operations. We treat the Fed's holdings before 2009 as fully anticipated through the scheduled-purchase control, so that its unanticipated demand before QE is zero by construction. We estimate the Fed's price elasticity and factor loadings from 2009 onward. Appendix B.1.2 gives the details.

[9] Household holdings of Treasuries are substantially driven by hedge funds domiciled in the Cayman Islands, while TIC data heavily undercounts Cayman-held Treasury securities (Barth et al., 2025). Such funds are predominantly managed by U.S. investors (Cayman Islands Monetary Authority, 2023). Enhanced Financial Accounts and Form PF data (2013 onwards) suggest that domestic and foreign hedge fund positions can account for more than 70 percent of household sector holdings of Treasury securities in recent years.

Treasury bonds outstanding in the last quarter A_{t-1}^{total}:

$$\Delta q_{i,t} \equiv \frac{F_{i,t}}{S_{i,t} A_{t-1}^{total}}.$$

As such, the sign of $\Delta q_{i,t}$ follows that of the dollar flows $F_{i,t}$, with positive numbers indicating net buying and negative numbers indicating net selling. Supply enters the system as negative demand, so that a positive net issuance over a quarter is registered as negative $\Delta q_{i,t}$, allowing us to jointly model supply and demand in this framework.

For most sectors, the size weight $S_{i,t}$ is defined following Gabaix, Koijen, Mainardi, Oh, and Yogo (2026):

$$S_{i,t} = \frac{|A_{i,t-1} + F_{i,t}/2|}{A_{t-1}^{total}}.$$

where $A_{i,t-1}$ denotes the market value of sector i's Treasury holdings as of time $t-1$. Defining $S_{i,t}$ in this way leads to

$$\Delta q_{i,t} = \frac{F_{i,t}}{|A_{i,t-1} + F_{i,t}/2|}.$$

This measure ensures robustness of the flow against small values of $A_{i,t-1}$. When prices do not change from quarter to quarter, $F_{i,t} = \Delta A_{i,t}$, and $\Delta q_{i,t}$ falls back to the canonical Davis and Haltiwanger (1992) growth rate.

Three sectors depart from this definition. The supply sector's size weight is set to one, so that net issuance is measured relative to the size of the market.[10] Households, a sector with a large hedge-fund component, and dealers behave in part as arbitrageurs in the Treasury market. Both sectors hold long and short Treasury positions and move between being net long and net short, so their net positions are not a meaningful measure of their size, and we fix their size weights at one as well. With this normalization, their estimated elasticities measure their flows relative to the total market rather than to their own holdings, and are therefore not comparable one for one with those of other sectors. Their contributions to the aggregate elasticity, however, are the product of size and elasticity, which the normalization leaves unchanged; these contributions keep their interpretation and remain comparable with those of other sectors.

Common factors. Common factors play two crucial roles in our analysis. First, they enable us to decompose and interpret price changes in the Treasury market. Second, as

[10]The optimal GIV estimator to be introduced in Section 3.3 does not require the size weights to sum to one for it to be efficient, as one can always apply rescaling.

we will outline later, our identification strategy relies on properly controlling for common factors to isolate idiosyncratic demand shocks.

We incorporate three broad sets of observable factors in the estimation, guided by both economic theory and their empirical explanatory power for yields. Details on the construction of these variables are relegated to Appendix B.1. The first set collects key macro-financial factors: (i) innovations to the implied volatility of the U.S. stock market, which serves as a primary indicator of global risk sentiment; (ii) quarterly revisions in futures-implied expectations of the next quarter's Fed funds rate, to account for expected short-rate movements; (iii) quarterly AR(1) innovations to the USD broad index; and (iv) quarterly AR(1) innovations to realized CPI inflation.[11] We do not include other asset prices that could be endogenously affected by idiosyncratic shocks in the Treasury market, such as observed credit spreads.

In addition, we account for policy influences on Treasury demand. Two key policy entities, the Federal Reserve and the Department of the Treasury (supply), regularly pre-announce their planned purchases and sales to provide transparency and forward guidance to market participants. To capture the anticipatory effects and potential front-running behaviors these announcements may induce, we include variables that can predict the policy sectors' flows in our common factors. Specifically, for the Fed, we control for purchases scheduled a quarter ahead.[12] For the Treasury, we include net issuance in the previous quarter to control for the anticipated component in Treasury supply. This approach allows us to isolate and measure the flow sensitivity due to deviations from scheduled purchases and sales when estimating coefficients for these policy sectors.

The final set of observable factors covers subjective expectations of key bond market participants. We extract unobserved factors from granular survey data. Specifically, we include four survey-based factors extracted from individual forecasts of the current-quarter Aaa corporate bond yield relative to the preceding quarter's realized yield, from the Survey of Professional Forecasters (SPF) panel data. Overall, the set of observable demand shifters explains more than 40% of variation of Treasury price changes over our sample period.[13]

[11] More broadly, accounting for macro factors is crucial to understanding Treasury price dynamics because they convey information not captured by the cross-section of yields (Ludvigson & Ng, 2009).

[12] Appendix B.1.2 discusses the construction of the scheduled purchase series.

[13] See Figure B.1 in Appendix B.1.2, which also illustrates the contribution of each factor to the fit by progressively adding macro-financial factors and reporting the R^2.

3.2 Identification Assumption

The main challenge in estimating the model is the endogeneity of Treasury prices to investors' allocation decisions. The standard solution is to use idiosyncratic demand shifters from one sector to instrument the price in another sector's demand equation, e.g., mutual-fund flows used to estimate stock and corporate bond elasticities (Lou, 2012; Chaudhary et al., 2026a). Such instruments are valid if the two sectors' idiosyncratic shifters are orthogonal, and relevant if the shocks are large enough to move prices. Our setting needs such an instrument for every sector and every period, and typically used instruments only cover particular sectors or short episodes. Moreover, in a market as large as Treasuries, a single sector's shocks are often too small to move prices, so the instruments lack relevance.

To address these challenges, we leverage our model's structure to extract idiosyncratic sector shocks. In our model, each sector's Treasury allocation depends on three components: (i) price (Δp_t), (ii) common demand shifters ($\boldsymbol{\eta}_t$), and (iii) idiosyncratic sector-specific shifters ($u_{i,t}$). The considerable heterogeneity in sectoral constraints and objectives makes these idiosyncratic drivers a significant source of variation in holdings. Our estimation strategy extracts these idiosyncratic sector-specific demand shifters from one sector and uses them as instruments for the yield in another sector.

Similar to standard instrumental variable approaches, our identification relies on idiosyncratic sector-specific drivers being orthogonal across sectors. For instance, selling Treasuries by a foreign country may be driven solely by their domestic concerns, and hence this can be used as an instrument for U.S. pension funds' demand. Formally, our identification assumption is based on conditional independence of idiosyncratic sector-specific shifters ($u_{i,t}$) across sectors:

Assumption 1. *$u_{i,t}$ is independent of $u_{j,t}$ for any $i \neq j$.*

Assumption 1 is also the identifying assumption of the GIV approach (Gabaix & Koijen, 2024), and our estimator extends the standard GIV estimator with important differences.[14]

Strict independence of the idiosyncratic shocks in Assumption 1 requires that we capture all common variation in the data; despite our extensive controls, some residual

[14]The key difference from the standard implementation is that our GIV extension *flexibly* accommodates heterogeneous price elasticities and factor loadings across investors without requiring explicit parameterization based on observable characteristics. In addition, our optimal GMM implementation more efficiently exploits information contained in idiosyncratic shocks. Appendix A.1 discusses the differences in more detail.

common factors may remain in $u_{i,t}$. In Section 3.4, we characterize the estimator's behavior under such violations, develop sharp diagnostics to quantify any resulting bias, and propose bias-reduction strategies. The bias rises only modestly with residual common variation, and our robustness exercises indicate that such variation is unlikely to be consequential.

3.3 The Optimal GIV Estimator

Based on Assumption 1, we propose an economically motivated optimal GMM estimator that weights investor shocks according to their market influence. This approach assigns greater importance to shocks from investors who have more significant impact on equilibrium yields, thereby enhancing the relevance of idiosyncratic demand shocks as the instrument for yields.

For ease of exposition, we focus on the case without common factors. Operationally, this can be achieved by first residualizing $\Delta q_{i,t}$ and Δp_t against the common factors and taking the residuals. We discuss the handling of common factors in Section 3.4. We also maintain the assumption for illustrative purposes that the elasticities are fully heterogeneous across sectors and constant across regimes. We hold the size weight $S_{i,t}$ constant to economize on notation. Appendix A.1 states the properties of our estimator in a fully general demand system with time-varying elasticities, factor loadings and sector sizes. We show how to incorporate external common factors in Appendix A.2.

Our identification Assumption 1 implies the following moment restrictions,

$$\mathbb{E}\left[u_{i,t}u_{j,t}\right] = 0, \quad i, j = 1, \ldots, N, i \neq j. \tag{3.2}$$

The set of moment conditions (3.2) yields an overidentified system of equations. More specifically, they place restrictions on the entire off-diagonal covariance matrix of idiosyncratic sector-specific shifters, giving us $N \times (N-1)/2$ moment conditions to identify just N sector-specific price elasticities. We would like to optimally weight the moment conditions. In theory, we could adopt efficient versions of GMM estimators off the shelf. However, a practical challenge is that the number of moment conditions $N \times (N-1)/2$ could far exceed the length of the time-series T in the sample. In such settings, statistical methods to find the optimal weighting matrix, such as two-step GMM, are known to perform poorly (Newey & Windmeijer, 2009; Han & Phillips, 2006).

Guided by the economics of our model, we derive an optimal weighting scheme for

the moment conditions. This optimal estimator assigns greater weight to moment conditions from entities with larger market shares. The rationale is straightforward: idiosyncratic shocks of entities with larger market shares generate greater price impact, making them more relevant as instruments. We refer to this size-weighted approach as the *optimal GIV estimator*.

Formally, the optimal GIV estimator $\hat{\boldsymbol{\zeta}}$ solves the following system of sample moment equations:

Definition 1. The sample moment conditions for the optimal GIV estimator are given by

$$\hat{\mathbb{E}}\left[\hat{u}_{i,t}\left(\hat{\zeta}_i\right)\underbrace{\sum_{j\neq i}S_j\hat{u}_{j,t}\left(\hat{\zeta}_j\right)}_{\hat{u}_{S(-i),t}\left(\hat{\boldsymbol{\zeta}}\right)}\right]\equiv\hat{\mathbb{E}}\left[\left(\Delta q_{i,t}+\hat{\zeta}_i\Delta p_t\right)\sum_{j\neq i}S_j\left(\Delta q_{j,t}+\hat{\zeta}_j\Delta p_t\right)\right]=0. \tag{3.3}$$

for all i, where $\hat{\mathbb{E}}$ denotes the sample mean across the time dimension.

Theorem 1, a special case of Theorem A.1, shows that the estimator is asymptotically efficient.

Theorem 1 (Asymptotic efficiency)**.** *Given the moment conditions (3.2), under regularity conditions laid out in Appendix A, the optimal GIV estimator $\hat{\boldsymbol{\zeta}}$ is consistent and asymptotically normal:*

$$\sqrt{T}\left(\hat{\boldsymbol{\zeta}}-\boldsymbol{\zeta}\right)\xrightarrow{d}\mathcal{N}\left(\mathbf{0},\mathbf{V}^{\zeta}\right),$$

for $T\to\infty$. Moreover, its asymptotic variance achieves the semi-parametric efficiency bound (Chamberlain, 1987), given by

$$\mathbf{V}^{\zeta}=\zeta_S^2\times\mathrm{Inv}\left(\begin{bmatrix}\frac{1}{\sigma_1^2}\sum_{i\neq1}S_i^2\sigma_i^2 & \cdots & S_1S_N\\ \vdots & \ddots & \vdots\\ S_1S_N & \cdots & \frac{1}{\sigma_N^2}\sum_{i\neq N}S_i^2\sigma_i^2\end{bmatrix}\right), \tag{3.4}$$

where $\zeta_S\equiv\sum_i S_i\zeta_i$ is the aggregate elasticity and $\sigma_i^2\equiv\mathrm{Var}(u_{i,t})$.

The expression (3.4) illustrates the sources of statistical power in the optimal GIV estimator. First, the standard error of each sector's elasticity, ζ_i, is scaled by the aggregate elasticity, ζ_S. This reflects that demand shocks have greater power when their price impact is larger. In the extreme case where the market is perfectly elastic ($\zeta_S\to\infty$), demand shocks cannot move prices at all, rendering the instruments irrelevant and causing stan-

dard errors to approach infinity. Second, the standard errors depend on the ratio of other sectors' size-weighted shock volatility, $\sum_{j\neq i} S_j^2\sigma_j^2$, to the sector's own shock volatility, σ_i^2. Higher volatility in the rest of the market creates larger exogenous price variations that can be used to instrument sector i's demand, enhancing instrument power. Conversely, when $\sum_{j\neq i} S_j^2\sigma_j^2 \to 0$, there is no exogenous price variation available to estimate sector i's price elasticity.

Instrumental-variable interpretation. Our estimator has a natural instrumental variable (IV) interpretation. For each entity i, the moment condition (3.3) can be rewritten as:

$$\hat{\zeta}_i = -\frac{\hat{\mathbb{E}}\left[\Delta q_{i,t}\hat{u}_{S(-i),t}\left(\hat{\boldsymbol{\zeta}}\right)\right]}{\hat{\mathbb{E}}\left[\Delta p_t\hat{u}_{S(-i),t}\left(\hat{\boldsymbol{\zeta}}\right)\right]}.$$

This is an IV estimator in which price changes are instrumented with the size-weighted idiosyncratic shocks of entities other than i. Maximizing the power of an IV estimator requires maximum instrument relevance, i.e., maximizing the covariance between price changes and the instrument. The market clearing condition of our model implies that the price change is an affine function of the size-weighted aggregation of idiosyncratic shocks, so that size weighting achieves maximal relevance. Appendix A.1 shows that the estimator can equivalently be written as an OLS regression of quantities on prices plus a correction for the bias from the sector's own price impact, a form that also gives an alternative, and sometimes more stable, algorithm.

3.4 Common Factors and the Robustness of Identification

Our identification method relies on the assumption that, after controlling for common factors, idiosyncratic shocks are orthogonal across sectors. The primary threat to identification is the presence of residual correlations in idiosyncratic shocks ($u_{i,t}$). We address this concern in a number of ways and provide a sharp diagnostic tool to assess the robustness of our estimates.

Our observable macro-financial factors, ranging from macroeconomic variables to market expectations about future yields, are comprehensive and explain a significant portion of Treasury price movements. If the observable demand shifters do not account for all common demand movements, our estimates will be biased. In Appendix A.3, we give propositions that sharply characterize the dependence of the magnitude of the bias on the residual comovement with price variation. Formally, the biased estimate of the aggregate

elasticity $\hat{\zeta}_S^b$ can be approximated by:

$$\hat{\zeta}_S^b \approx \zeta_S \sqrt{1 - \mathcal{R}^2} \tag{3.5}$$

where $\mathcal{R}^2$ represents the share of the remaining price variance explained by common factors not controlled for, i.e., $\mathcal{R}^2 = \frac{\text{Var(hidden factors)}}{\text{Var}(\Delta p_t^{resid.})}$. The term $1 - \mathcal{R}^2$ captures the share of remaining price variance explained by true idiosyncratic shocks.[15] The higher the share of common comovement not controlled for, the stronger the downward bias in the estimated aggregate elasticity relative to the truth.

Based on Equation (3.5), it is likely that the impact of residual comovement on our estimates would be relatively modest. For instance, if hidden factors explain 25% of the remaining price variance ($\mathcal{R}^2 = 0.25$), our estimated elasticity would be biased downward by only about 13% ($\sqrt{1 - 0.25} \approx 0.87$). To generate a 50% bias in our aggregate estimates, hidden factors would need to explain 75% of the remaining price variance. Given our comprehensive set of controls and the factors extracted by principal component analysis (PCA), such a high level of unexplained common comovement seems unlikely.

To test this intuition directly, we develop an alternative set of moment conditions that take into account the factor structure for residuals (Appendix B.2). We discuss identifiability, and develop a method to jointly estimate "internal PCA" from holdings data along with other parameters of the model. Internally extracted factors provide little explanatory power for Treasury price changes beyond what observable controls provide: projecting price changes on one to three internal principal components gives an R^2 of at most 2.8%, with an adjusted R^2 of 0.9%, so the bias from residual comovement with prices is small by (3.5). Our key takeaways also remain unchanged when the internal components replace the survey-based factors in estimation (Appendix B.2, Figure B.4).

For sector-specific estimates, strategic behavior among sophisticated investors also raises valid concern. For example, fast-moving investors might front-run a large but slow-moving investor's positive demand shock. Such strategic interactions would violate Assumption 1 and bias our estimates. To address this concern, we leverage the over-identifying nature of our GMM procedure to implement a leave-one-out estimation approach, by excluding one sector at a time from the set of moment conditions to test

[15] The bias formula shares the same spirit as the price impact bound developed by van der Beck, Bretscher, and Fu (2025), which stresses the tension between heterogeneity across investors $\sqrt{1-\rho}$ and the market elasticity when inferring from the price and flow volatility. While both formulas involve R-squared, a subtle difference is that here $\mathcal{R}^2$ is the R-squared of *remaining price changes* explained by common factors, while the homogeneity ρ for the price impact bound is the R-squared of total demand shocks explained by cross-investor average.

the sensitivity of our results to particular sectors. The estimates are robust to excluding any single sector: the time-average aggregate elasticity ranges from 0.97 to 1.04 across the leave-one-out fits, against 1.01 in the full fit, and its decline across 2009Q1 keeps its sign and size in every one of them (Appendix B.2, Figures B.2 and B.3). Strategic interactions are therefore not a significant concern in our setting.[16]

Appendix B.2 collects these checks and two further diagnostics. The estimated idiosyncratic shocks $\hat{u}_{i,t}$ are close to uncorrelated across sectors (Figure B.5). China's extracted demand shocks line up with narrative evidence on its reserve management (Figure B.6). The aggregate shock $\hat{u}_{S,t}$ moves yields across the whole term structure in line with a duration-based extrapolation of the price impact on the market portfolio (Figure B.7), so our demand shocks mostly shift the level of the yield curve, consistent with our focus on the overall level of yields.

4 Shifting patterns of Treasury market liquidity provision

A deep and liquid Treasury market can accommodate large shifts in investor demand without large price impact. We measure this capacity by the market's aggregate price elasticity ζ_S, the size-weighted sum of investors' price elasticities, and by its reciprocal, the macro multiplier $M = 1/\zeta_S$: a demand shift equal to 1% of the market moves prices by M percent. This section documents how this capacity has evolved since 1986 and which investors provide it, against the backdrop of shifting investor composition (Figure 1).

4.1 The market's liquidity has declined moderately

Table 1 reports the aggregate price elasticity of the Treasury market before and after the global financial crisis, in panel (a), and the price elasticities of the sectors that make it up, in panel (b). The crisis brought profound structural changes across many sectors of the Treasury market. Among the most significant was the Federal Reserve's emergence as an active participant: it began large-scale purchases of Treasuries in March 2009, so we date the regime shift to 2009Q1. Every sector's elasticity, factor loadings and intercept differ before and after 2009Q1, so the two columns are two separate fits of the demand system on either side of the financial crisis. Before the crisis, a positive 1% demand shock raised Treasury prices by 0.88%, or equivalently, lowered yields by 16.0 basis points; since the

[16] Appendix A.1 points to further discussion of conceptual issues in identification, including handling missing sectors and measurement errors. For sectors with potential measurement errors in holdings data, we can adopt the same leave-one-out approach to exclude moment conditions involving such sectors, and check the robustness of our estimates.

crisis, the same shock raises prices by 1.22% and lowers yields by 22.1 basis points. Compared with estimates for other asset classes in the literature, macro multipliers of 0.88 to 1.22 suggest that the U.S. Treasury market absorbs demand shocks far more effectively than the U.S. equity market, which has a macro multiplier closer to 5 (Gabaix & Koijen, 2022), and the U.S. corporate bond market, which has a portfolio-level multiplier of 3.5 (Chaudhary et al., 2026a). These differences in macro multipliers across asset classes are consistent with the principle that multipliers increase when outside assets are poor substitutes, which makes it harder for arbitrageurs to hedge risks when countering demand shocks (Chaudhary et al., 2026a).

The literature has also estimated the price impact of various events on Treasury yields, including foreign FX interventions, quantitative easing, Treasury repurchase operations, and other policy measures. These studies provide reduced-form price impact estimates around events associated with demand/supply shocks. By making a few assumptions on the implicit size of the associated quantity changes, we can map these price-impact estimates to multipliers to obtain useful benchmarks. Appendix B.3.1 shows that our Treasury macro-multiplier estimates align with the magnitudes implied by existing price-impact studies.

The aggregate elasticity thus fell significantly across the financial crisis, from 1.14 to 0.82. The decline is moderate, however, given how completely the investor base turned over during the same period (Figure 1). The next section shows why: private investors withdrew from liquidity provision, and the Federal Reserve offset much of what they withdrew.

4.2 Private investors have retreated and the Fed has filled the gap

A sector's price elasticity of demand reflects its willingness to stand on the other side of the market when demand and supply shocks move Treasury prices. In panel (b) of Table 1, we set out to explain the moderate shift in aggregate elasticity by examining individual sectors. Figure 2 further quantifies each sector's contribution to the aggregate shift. The pre-2009 column establishes that there is substantial heterogeneity in the liquidity-provision capacity at the sector level, with banks by far the most elastic sector. ETFs, which enter the sample as a separate sector in 2009, have a negative point estimate, potentially reflecting return-chasing behavior by retail end-investors; due to their small market share, they contribute very little to the aggregate elasticity. Meanwhile, on the supply side, we estimate an elasticity close to zero. This result reflects the Treasury's commitment to a "regular and predictable" debt management approach. As we control

Table 1: **Price elasticity of the Treasury market and its sectors before and after 2009**

	(1) Pre-2009	(2) Post-2009	(2)−(1)
(a) Market aggregate			
Aggregate elasticity ζ_S	1.14***	0.82***	−0.32***
	[0.95, 1.33]	[0.69, 0.96]	[−0.55, −0.08]
Macro multiplier $M = 1/\zeta_S$	0.88***	1.22***	0.34***
	[0.73, 1.02]	[1.02, 1.42]	[0.09, 0.58]
Yield move from a 1% sale (bp)	16.0***	22.1***	6.1***
	[13.3, 18.6]	[18.5, 25.7]	[1.6, 10.6]
(b) Price elasticity by sector			
Households†	0.41***	0.25***	−0.16
	[0.23, 0.60]	[0.14, 0.36]	[−0.38, 0.05]
Rest of world	0.52***	0.09	−0.43***
	[0.32, 0.72]	[−0.06, 0.24]	[−0.68, −0.18]
Banks	1.69***	1.12***	−0.57**
	[1.32, 2.05]	[0.86, 1.37]	[−1.02, −0.12]
Dealers†	0.09*	0.03	−0.06
	[0.00, 0.18]	[−0.04, 0.09]	[−0.17, 0.05]
Fed		1.21***	
		[0.98, 1.43]	
Pension funds	0.54***	0.21**	−0.33**
	[0.28, 0.80]	[0.03, 0.39]	[−0.64, −0.01]
Other	−0.01	0.54***	0.55***
	[−0.28, 0.27]	[0.35, 0.73]	[0.21, 0.88]
Mutual funds	0.69***	0.43***	−0.25
	[0.23, 1.14]	[0.11, 0.76]	[−0.81, 0.31]
Insurance	0.47***	0.41***	−0.06
	[0.21, 0.73]	[0.23, 0.59]	[−0.37, 0.26]
ETFs		−0.20	
		[−0.54, 0.13]	
Supply†	0.13***	0.08***	−0.06
	[0.05, 0.21]	[0.02, 0.13]	[−0.15, 0.04]

Note: This table reports estimates from the specification in which every sector's price elasticity, factor loadings, and intercept differ before and after 2009Q1. Panel (a) reports the aggregate price elasticity $\zeta_S = \sum_i S_i \zeta_i$, the macro multiplier $M = 1/\zeta_S$, and the yield change implied by a demand shift of 1% of the market at a 5.5-year duration, around the sample average. Panel (b) reports the sector price elasticities ζ_i. Sectors marked † have unit size weights, so their coefficients measure demand responses relative to the total market; the other sectors' coefficients are relative to their own holdings. The Federal Reserve's pre-2009 purchases are treated as fully anticipated through the scheduled-purchase control, so its pre-2009 coefficients are not estimated and are left blank. ETFs enter the sample in 2009Q1. The sample is 1986Q1–2023Q4. 95% confidence intervals, from equation (A.5), are in brackets below estimates. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

for the preannounced supply schedules, there is little room for the supply to further respond to price changes.

The change column of panel (b) shows a broad-based decline in price elasticities across sectors. Among the sectors with statistically significant changes, banks' price elasticity fell by 34%, and pension funds' price elasticity fell by 61%. For banks, leverage requirements make expanding Treasury positions costly in terms of balance-sheet capacity and may limit their willingness to absorb additional securities when prices fall (Cochran, Infante, Petrasek, Saravay, & Tian, 2023). Liquidity regulation may also make the adjustment of Treasury holdings less flexible by tying Treasury demand to required liquidity buffers (Stein, 2013). For pension funds, the shift from defined-benefit to defined-contribution plans, greater delegation to investment funds and the associated changes in investment mandates may have reduced their role as direct, discretionary Treasury buyers (Ding, Fang, Hardy, & Lewis, 2026; Andonov, Jansen, & Rauh, 2026).

The main sector whose demand becomes more elastic is the Federal Reserve. Before the financial crisis, changes in the Fed's Treasury notes and bonds holdings are typically associated with its routine operational needs. We thus assume that all changes in the holdings in this period are perfectly anticipated and control for the Fed's scheduled purchases throughout, so that its Treasury demand would be perfectly inelastic.[17] After the crisis, however, as the Fed initiates large-scale asset purchases, its demand becomes more elastic, with an estimated elasticity of 1.21 (Table 1, panel (b)). Our analysis reveals the Fed is price elastic, purchasing additional Treasury bonds when the Treasury market experiences downward price pressures. This behavior aligns with the goal of Treasury market stabilization—one of the stated objectives of QE operations (Gagnon et al., 2011a).[18]

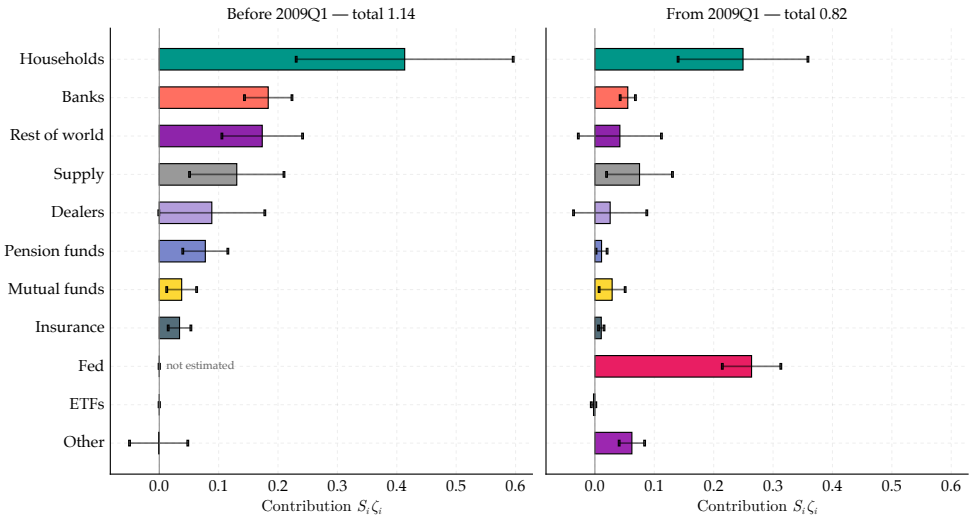
In Figure 2, Panel (a), we report the contributions of different sectors to the aggregate elasticity before and after 2009 by multiplying their respective size weights with the sector-level elasticities. Non-Fed investors' contributions to the aggregate elasticity fell by 0.52 across 2009Q1, led by households (-0.16), the rest of the world (-0.13), banks (-0.13), pension funds (-0.07), and dealers (-0.06); Treasury supply contributed a further -0.06. The household sector, whose holdings are mostly driven by hedge funds, remains the one with the largest relative contribution other than the Fed, reflecting its role as a nimble liquidity provider attributable to a looser regulatory constraint. The Fed-

[17] Alternative approaches—treating all changes as unanticipated or using an AR(1) process to control for expected changes—all yield near-zero elasticity estimates for the Fed in the pre-crisis period. As a byproduct, the pre-crisis estimates of the Federal Reserve's Treasury demand loading on all common factors except the anticipated purchases are almost identical to zero. See Appendix B.1.2 for more details.

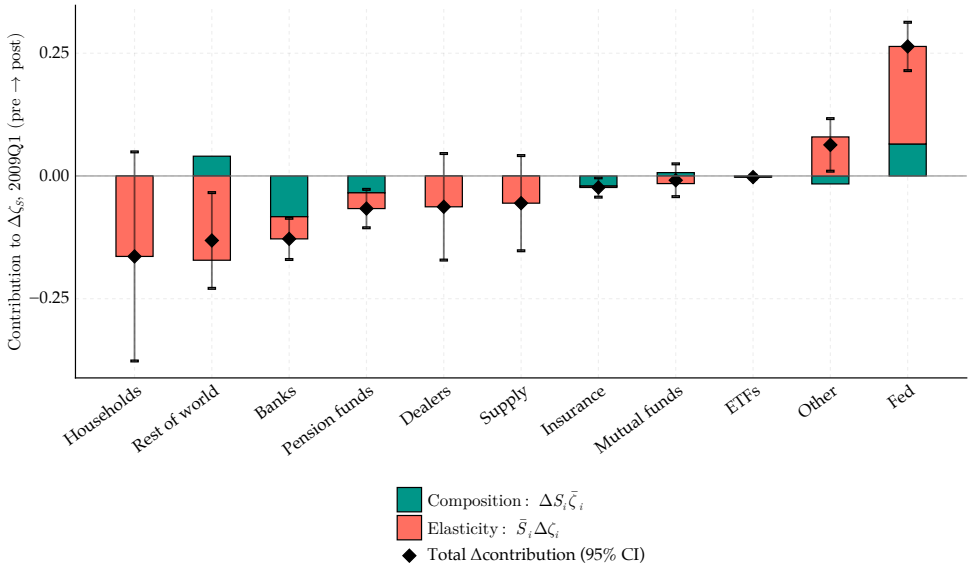
[18] The phrase "the smooth functioning of the Treasury market" appears frequently in Federal Open Market Committee minutes. For example, a 2020 New York Fed announcement stated: "The Desk stands ready to adjust the size and composition of its purchase operations as appropriate to support the smooth functioning of the Treasury market." Another contemporaneous announcement specified that "the Desk will conduct purchase...subject to reasonable prices."

Figure 2: **Contributions to aggregate elasticity before and after 2009**

(a) Contribution levels



(b) Change across 2009Q1: composition and elasticity



Note: This figure decomposes the aggregate elasticity of Table 1 into sector contributions. Panel (a) plots each sector's contribution $S_i\zeta_i$ to the aggregate elasticity, with 95% confidence intervals from equation (A.5); each regime's bars sum to ζ_S. Panel (b) splits the change in each contribution into a composition effect, $\Delta S_i\,\bar{\zeta}_i$, and an elasticity effect at the average size weight, $\bar{S}_i\,\Delta\zeta_i$. The two stacked components sum to the total change, marked by the diamond with its 95% confidence interval.

eral Reserve, on the other hand, becomes the biggest positive contributor to aggregate elasticity, offsetting about half of the negative impact from other investors and supplying around a third of the post-2009 aggregate elasticity. Holding other investors' behavior

and size weights fixed, removing the Fed's post-crisis contribution would increase the price impact of a sale equal to 1% of the market from 1.22% to about 1.79%, or from 22 to about 33 basis points in yields—roughly twice its pre-crisis level.

In Panel (b), we split the changes in the sectors' contributions into components driven by changes in their size weights that capture composition effect, and changes in their elasticities. Among the sectors showing significant elasticity decline, banks and pension funds experience a decline in both size and elasticity, while foreign investors' substantial negative contribution mainly comes from their demand becoming more price inelastic.

Taken together, the Treasury market's liquidity now rests partly on the central bank rather than on private and foreign investors. If the retreat of these investors is permanent, the market's ability to accommodate investor demand shifts could decline further during future policy tightening episodes when the Fed reduces its Treasury holdings.

4.3 Foreign holdings no longer translate into liquidity provision

Foreign investors are the clearest case of the retreat documented in Section 4.2. They held nearly two-thirds of the market in 2008 and still about a third in 2023, yet their contribution to the aggregate elasticity fell by 0.13 across 2009Q1, almost entirely because their demand became less price-sensitive rather than because their holdings shrank (Figure 2, panel (b)). On average, foreign investors' price elasticity declines from 0.52 to 0.09, with the latter not significantly different from zero (Table 1, panel (b), reproduced as panel (a) of Table 2). In order to understand the underlying drivers of this change, we keep the 2009 regime shift and allow the price elasticity within the foreign sector to vary across major country groups. We split countries into five groups: China, Japan, Europe, Americas, and a residual "other countries" category.[19]

Panel (b) of Table 2 reports region-specific elasticity shifts. Overall, allowing heterogeneous price elasticities across country groups barely changes the aggregate market elasticity estimates. Japan, China, and Europe all show significant elasticity declines. Their post-2009 price elasticities become statistically indistinguishable from zero. China's estimated price elasticity falls by approximately 82%, from 2.71 to 0.50, while Japan's falls from 0.59 to -0.05 and Europe's from 0.48 to -0.07. Overall, foreign investors now play a substantially smaller role in absorbing demand shocks in the Treasury market. Potential

[19]The Europe (Denmark, Finland, Germany, the Netherlands, Spain, Sweden, Switzerland, and the United Kingdom) and Americas (Canada, Brazil, Chile, and Colombia) groups contain countries that consistently appear in TIC data with substantial positions. Other countries in these continents are grouped in the residual category.

explanations of this retreat include a lower precautionary demand overall, more diversified use of foreign reserves, as well as geoeconomic tensions.

Table 2: **Regime shifts for price elasticities of foreign investors, before and after 2009**

	1986Q1–2008Q4			2009Q1–2023Q4			Change	
Investor	ζ	Contrib.	Share (%)	ζ	Contrib.	Share (%)	$\Delta\zeta$	ΔContrib.
(a) Common foreign elasticity								
Rest of world	0.52^{***}	0.17^{***}	15.24	0.09	0.04	5.11	-0.43^{***}	-0.13^{***}
	[0.32, 0.72]	[0.11, 0.24]		[−0.06, 0.24]	[−0.03, 0.11]		[−0.68, −0.18]	[−0.23, −0.03]
Market aggregate	1.14^{***}			0.82^{***}			-0.32^{***}	
	[0.95, 1.33]			[0.69, 0.96]			[−0.55, −0.08]	
(b) Regional elasticities								
China	2.71^{***}	0.10^{***}	8.39	0.50	0.05	6.46	-2.21^{*}	−0.04
	[1.04, 4.37]	[0.04, 0.16]		[−1.15, 2.14]	[−0.12, 0.23]		[−4.55, 0.13]	[−0.23, 0.14]
Japan	0.59^{***}	0.06^{***}	4.98	−0.05	0.00	−0.51	-0.64^{**}	-0.06^{***}
	[0.20, 0.99]	[0.02, 0.10]		[−0.33, 0.24]	[−0.03, 0.02]		[−1.12, −0.15]	[−0.11, −0.01]
Europe	0.48^{**}	0.03^{**}	2.71	−0.07	0.00	−0.43	-0.55^{**}	-0.03^{**}
	[0.07, 0.89]	[0.00, 0.06]		[−0.37, 0.23]	[−0.02, 0.01]		[−1.06, −0.04]	[−0.07, 0.00]
Americas	0.46	0.01	0.48	0.33	0.01	1.23	−0.14	0.00
	[−0.42, 1.35]	[−0.01, 0.02]		[−0.31, 0.97]	[−0.01, 0.03]		[−1.23, 0.95]	[−0.02, 0.03]
Other countries	0.37^{**}	0.05^{**}	3.98	0.24^{*}	0.04^{*}	5.36	−0.13	0.00
	[0.02, 0.71]	[0.00, 0.09]		[−0.02, 0.50]	[0.00, 0.09]		[−0.56, 0.30]	[−0.07, 0.06]
Rest of world	0.71^{***}	0.24^{***}	20.55	0.21	0.10	12.11	-0.49^{**}	−0.14
	[0.45, 0.97]	[0.15, 0.32]		[−0.18, 0.61]	[−0.08, 0.28]		[−0.96, −0.02]	[−0.34, 0.07]
Market aggregate	1.15^{***}			0.83^{***}			-0.33^{**}	
	[0.96, 1.35]			[0.66, 1.00]			[−0.58, −0.07]	

Note: This table reports foreign price elasticities ζ_i from the specification of Table 1. Panel (a) reproduces the foreign sector's elasticity from Table 1; panel (b) re-estimates the same specification with separate elasticities for five country groups: China, Japan, Europe, Americas, and Other countries. Contrib. is the contribution $S_i\zeta_i$, averaged over the regime's quarters, and Share is the contribution divided by the aggregate elasticity ζ_S. In panel (b), Rest of world sums the five groups' contributions and reports their size-weighted elasticity. Market aggregate is ζ_S. Changes are post-2009 minus pre-2009. The sample is 1986Q1–2023Q4. 95% confidence intervals, from equation (A.5), are in brackets below estimates. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Foreign demand matters less for yields since the crisis. We quantify the contribution of each country to Treasury yield changes over time using the regional specification in Table 2, panel (b). By equation (2.4), group i contributes

$$\Delta p_{i,t} = \frac{1}{\zeta_{S,t}} S_{i,t} \left(\overline{\Delta q}_i + \boldsymbol{\lambda}_{i,t} \boldsymbol{\eta}_t + u_{i,t} \right)$$

to the price change: its demand shift at constant prices, scaled by the macro multiplier. The contributions sum to the observed price change. We convert them to yield changes at the lagged Treasury duration, so a negative contribution means the group's demand lowered yields. Figure 3 shows foreign countries' contribution to Treasury yield changes,

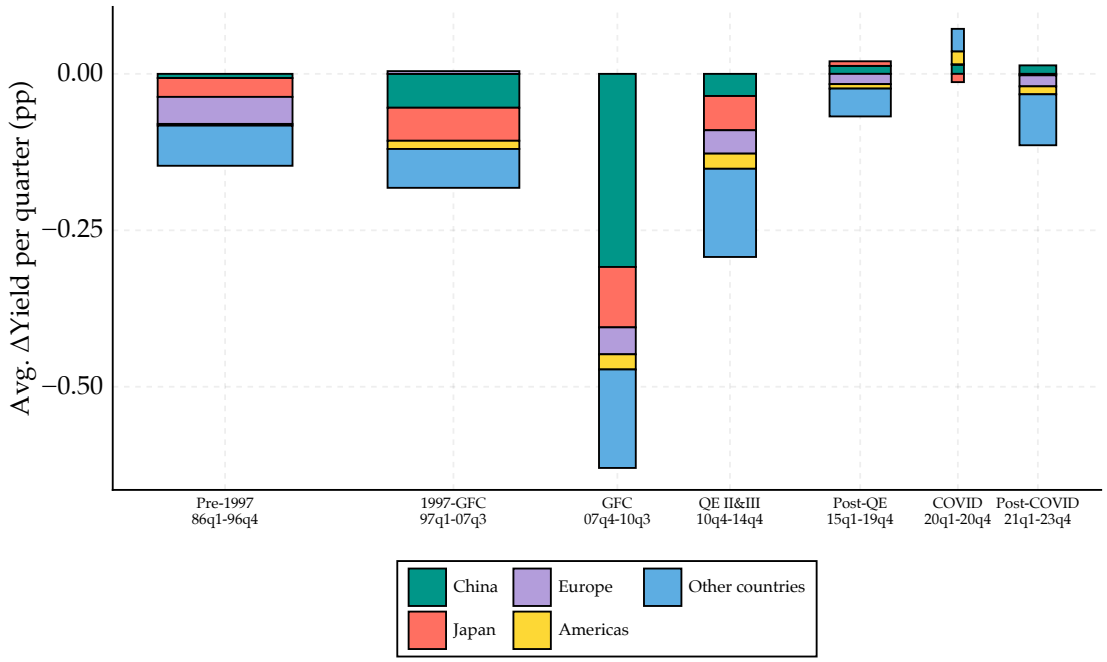
and Appendix Figure B.9 presents all investor groups. Foreign investors were the dominant driver of Treasury yields before the global financial crisis, contributing an average yield reduction of 28.9 basis points per quarter over 2003Q4–2007Q3. Their contribution peaked during the crisis, at 63.0 basis points per quarter over 2007Q4–2010Q3, and has waned since. Between 2003Q4 and 2007Q3, China, Europe, and Japan accounted for 56% of the foreign sector's impact on yields and lowered them by 0.65 percentage points per year; China alone depressed yields by a cumulative 1.4 percentage points. Chinese demand still lowered yields by about 31 basis points per quarter during the crisis (2007Q4–2010Q3), but its contribution all but disappeared after 2015, and foreign investors' gross contribution over 2020 was a 23 basis point increase in yields.[20] These estimates speak to the global savings glut debate: current-account-surplus countries channeled nearly 3 trillion USD into Treasuries during the 2000s, and the associated compression of long-term yields figured in debates surrounding Greenspan's "conundrum" and Bernanke's "savings glut" hypothesis (Greenspan, 2005; Bernanke, 2005; Caballero, 2006; Mendoza, Quadrini, & Ríos-Rull, 2009). Our decomposition shows that this compression was concentrated in the first decade of the 2000s and has not recurred.

The shrinking market share and diminished price impact of foreign investors also limit the disruption that unilateral foreign selling could cause today, say by a major holder such as China. Two forces have reduced China's footprint in the market: (i) rapid Treasury issuance since the crisis has lowered every foreign holder's share of the market; and (ii) China itself has withdrawn, cutting its holdings by about 40%, from $1.31 trillion in November 2013 to $0.80 trillion in December 2023. Consequently, China's share of Treasury notes and bonds fell from a 2010Q3 peak of 17.8% to 3.9% at the end of 2023. To quantify what this means for the most discussed scenario—a full liquidation of China's Treasury holdings within a single quarter[21]—we combine the regional specification's macro multiplier of 1.29 at end-2023 with China's 3.9% share of notes and bonds and a duration of 6.47 years: such a liquidation would raise Treasury yields by approximately 77 basis points, about 1.60 times the 48-basis-point standard deviation of quarterly yield changes over 1986Q1–2023Q4. The same liquidation at the end of 2008, when China held 17.1% of the market and the multiplier stood at 0.72, would have raised yields by 2.2 percentage points—a significant shock at the height of the crisis.

[20] China's reserve sales in 2015, prompted by depreciation pressure after the end of QE, had minimal price impact compared with its pre-crisis purchases.

[21] Reportedly proposed but rejected during the global financial crisis, the scenario has been characterized as China's economic "nuclear option" in trade disputes (Sorkin, 2018; Paulson, 2013).

Figure 3: **Important foreign holders' contributions to yields, percentage points per quarter**



Note: This figure plots the average quarterly contribution to Treasury yield changes, in percentage points, of the five foreign country groups in Table 2, panel (b), averaged within seven episodes; bar widths reflect episode lengths, and GFC denotes the global financial crisis. A group's contribution is $\Delta p_{i,t}$ in equation (2.4), converted to a yield change at the lagged Treasury duration; negative values mean the group's demand lowered yields.

5 Investors and the evolving safe-haven property

Alongside liquidity, the safe-haven property underpins the central role of the Treasury market: Treasuries tend to appreciate when global risk rises, most visibly during the global financial crisis, and this insurance is part of what makes them the natural home for the world's reserves and precautionary balances. Identifying the investors behind this response helps discipline models in which countercyclical dollar or Treasury demand shapes international risk sharing (Kekre & Lenel, 2024; Devereux et al., 2023). Linking the property to financial intermediation and market structure also speaks to a more practical question; namely, whether changes in the investor base can strengthen or erode the ability of Treasuries to insure global investors in future periods of stress.

We measure the safe-haven property as the response of Treasury prices to an innovation in uncertainty. The uncertainty measure is given by the implied U.S. stock-market volatility (VIX), extended back to 1986 using the VXO index.[22] By the pricing equation

[22]See Appendix B.1.2 for more detail. We directly use innovations in implied stock-market volatility as

(2.3), this response, β_t^{unc}, is the aggregate uncertainty loading divided by the aggregate elasticity:

$$\beta_t^{unc} = \frac{\lambda_{S,t}^{unc}}{\zeta_{S,t}} = \frac{1}{\zeta_{S,t}} \sum_i S_{i,t} \lambda_{i,t}^{unc}.$$

Investor i therefore contributes $S_{i,t}\lambda_{i,t}^{unc}$ to the aggregate demand loading and $S_{i,t}\lambda_{i,t}^{unc}/\zeta_{S,t}$ to the price response. Our estimates identify which investors' demand rises when uncertainty jumps, which investors' demand falls, and how these responses have changed over time.

To capture each investor's full demand response, we report *total* factor loadings. The total loadings include the influence that runs through responses of the other macro-financial factors to innovations to uncertainty. Increases in uncertainty may strengthen expectations of monetary policy easing, which in turn raises investors' demand. Because we control for short-rate expectations, the *direct* loading—the coefficient on uncertainty itself, holding the other factors fixed—excludes this channel, whereas the total loading includes it. In Appendix B.3.3, we formally define both loadings, and show that our takeaway remains robust when we use direct loadings to measure investors' response to uncertainty.

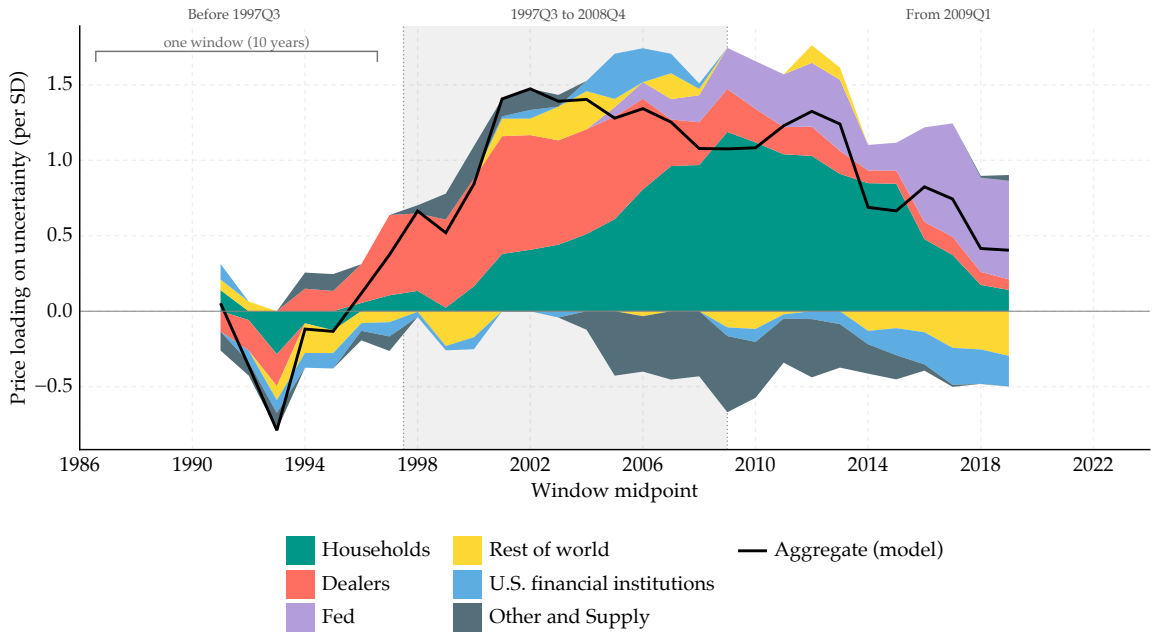
We first trace the safe-haven property over time by re-estimating the demand system in ten-year rolling windows. Figure 4 stacks the resulting contributions by investor group, with each window plotted at its midpoint; the black line is the model's aggregate price loading, the sum of the plotted contributions. It closely tracks the ten-year rolling regression of Treasury price changes on the uncertainty innovation (Appendix Figure B.10). The three regimes stand out. In windows centered before 1997, the price loading is close to zero or negative: Treasury prices did not rise when uncertainty jumped. It climbs through the late 1990s, peaks at about 1.5 in windows centered around 2002, and stays between 1.1 and 1.4 in the windows that span the financial crisis. It then declines once the windows are centered after 2013, and it ends the sample at 0.4, less than a third of its peak.

To test this profile formally, we split the sample into three regimes, 1986Q1–1997Q2, 1997Q3–2008Q4 and 2009Q1–2023Q4.[23] We re-estimate the demand system with every sector's price elasticity, uncertainty loading, and intercept allowed to differ across the

a barometer of global risk sentiment. Another related measure is stock–bond correlation, but it reflects a mixture of growth, monetary policy, and financial shocks (Cieslak & Pang, 2021; Antolín-Díaz, 2026).

[23] We place the first regime shift at 1997Q3, near the rise in the rolling price loading on uncertainty, and the second at 2009Q1, the same regime shift used in the liquidity analysis of Section 4. Appendix B.3.3 assesses the robustness of this timing choice using price regressions and demand-system estimates with alternative dates for the first shift, holding the second fixed at 2009Q1.

Figure 4: **Ten-year rolling contributions to the safe haven, by investor group**



Note: This figure plots 10-year rolling-window estimates of each group's contribution to the price loading on uncertainty, $\sum_i S_{i,t}\lambda_{i,t}^{unc}/\zeta_{S,t}$ over the group's sectors, using total loadings. U.S. financial institutions are banks, insurers, pension funds, mutual funds, and ETFs; Other and Supply are the Other sector and Treasury supply. For numerical stability, the household and dealer elasticities are fixed at their estimates from a single-regime specification, and all of the Federal Reserve's coefficients at their full-sample estimates before and after 2009Q1. The solid black line is the model aggregate, the sum of the contributions. Windows are plotted at their midpoints; dotted vertical lines mark the regime shifts at 1997Q3 and 2009Q1, and the shaded band marks windows whose midpoint falls between them.

three regimes, while the other factor loadings remain pooled. Table 3 reports each group's loading and its contribution to the aggregate.

Together, the rolling profile and the regime table establish three facts, which the rest of this section documents in turn: (i) the safe haven emerged in the late 1990s and has faded; (ii) foreign investors did not produce it; and (iii) dealers drove both its rise and its decline.

5.1 The safe haven emerged in the late 1990s and has faded since 2012

The aggregate row of Table 3 summarizes the three regimes. Before 1997Q3, the aggregate demand response to a one-standard-deviation rise in uncertainty is -0.35 and not significantly different from zero; over 1997Q3–2008Q4 it is 2.14, significant at the 1% level; and since 2009 it is 0.43, again indistinguishable from zero. Dividing by the regime's aggregate elasticity converts these demand responses into price responses: in the middle regime, a

Table 3: **Uncertainty loadings and contributions across three regimes**

	1986Q1–1997Q2		1997Q3–2008Q4		2009Q1–2023Q4	
	Loading	Contribution	Loading	Contribution	Loading	Contribution
Aggregate	−0.35	—	2.14***	—	0.43	—
	[−1.77, 1.07]		[1.02, 3.26]		[−0.70, 1.55]	
(a) Contributions to the aggregate						
Rest of world	0.28	0.06	0.10	0.04	−0.80	−0.37
	[−1.09, 1.64]	[−0.22, 0.34]	[−1.22, 1.42]	[−0.57, 0.66]	[−1.77, 0.17]	[−0.82, 0.08]
Households†	−0.02	−0.02	0.81**	0.81**	0.40**	0.40**
	[−0.63, 0.58]	[−0.63, 0.58]	[0.16, 1.47]	[0.16, 1.47]	[0.06, 0.74]	[0.06, 0.74]
Dealers†	−0.10	−0.10	1.31***	1.31***	0.03	0.03
	[−0.50, 0.31]	[−0.50, 0.31]	[0.88, 1.74]	[0.88, 1.74]	[−0.22, 0.29]	[−0.22, 0.29]
Remaining domestic	−0.20	−0.14	−0.20	−0.09	−0.42	−0.14
	[−1.33, 0.92]	[−0.91, 0.63]	[−1.17, 0.77]	[−0.50, 0.33]	[−1.07, 0.22]	[−0.34, 0.07]
Supply†	−0.15	−0.15	0.06	0.06	0.02	0.02
	[−0.48, 0.18]	[−0.48, 0.18]	[−0.30, 0.43]	[−0.30, 0.43]	[−0.19, 0.23]	[−0.19, 0.23]
Fed					2.18	0.48
					[−0.61, 4.97]	[−0.13, 1.09]
(b) Rest of world vs. domestic private						
Domestic private	−0.35	−0.26	3.94***	2.03***	0.80	0.30
	[−2.05, 1.36]	[−1.54, 1.02]	[2.20, 5.68]	[1.13, 2.93]	[−0.56, 2.17]	[−0.21, 0.81]
Difference	0.62	0.32	−3.84***	−1.99***	−1.60*	−0.67**
	[−1.27, 2.51]	[−0.92, 1.55]	[−6.05, −1.63]	[−3.09, −0.89]	[−3.25, 0.04]	[−1.34, 0.00]

Note: This table reports the total uncertainty loadings λ_i^{unc}, from the specification in which every sector's price elasticity, uncertainty loading, and intercept differ across the three regimes; the other factor loadings are constant. Contribution is $S_i\lambda_i^{unc}$, averaged over the regime's quarters; the contributions in panel (a) sum to the aggregate demand loading λ_S^{unc} in the Aggregate row. Sectors marked † have unit size weights, so their coefficients measure demand responses relative to the total market; the other sectors' coefficients are relative to their own holdings. Remaining domestic combines banks, insurers, pension funds, mutual funds, ETFs, and Other; Domestic private combines households, dealers, and remaining domestic. The Domestic private loading is its contribution normalized by the domestic private sectors' total share of Treasury holdings. The Difference row is the rest of the world minus domestic private. The Federal Reserve's pre-2009 purchases are treated as fully anticipated through the scheduled-purchase control, so its pre-2009 coefficients are not estimated. The sample is 1986Q1–2023Q4. 95% confidence intervals, computed as in Appendix B.3.3, are in brackets below estimates. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Appendix Table B.2 reports direct loadings.

one-standard-deviation rise in uncertainty raised Treasury prices by about 1.5%, equivalent to a fall in yields of roughly 27 basis points at an average duration of around 5.5 years, whereas in the other two regimes the price response is statistically zero. Within the post-2008 regime, the rolling estimates in Figure 4 show that the safe haven persisted until about 2012 and then declined; by the end of the sample, the price response to uncertainty is less than a third of its peak. The rise and fall we document echoes the change in the stock–bond correlation and the concerns raised during the COVID-19 crisis about Treasuries' waning safety (Baele, Bekaert, & Inghelbrecht, 2010; Campbell et al., 2020; Antolín-Díaz, 2026; He et al., 2022; Vissing-Jorgensen, 2021).

5.2 Foreign investors did not make Treasuries a safe haven

We next examine which investors have shaped the safe-haven property of Treasuries over time. Foreign investors are a natural candidate emphasized in the literature. The *flight-to-safety view* holds that foreign investors increase their demand for Treasuries during global financial tightening, supporting the countercyclical behavior of Treasury prices (Jiang et al., 2021, 2024). This view is central to models linking the dollar's special status to the United States' external funding advantage. The *precautionary saving view* holds instead that foreign investors accumulate Treasuries in normal times and draw them down to raise liquidity when international financial markets are disrupted (e.g., Eisenbach & Phelan, 2026; Bianchi et al., 2018; Devereux et al., 2023). Under this view, foreign investors benefit from the Treasury's hedging value during global downturns, but they do not lower Treasury yields by raising their demand in those downturns.

Observed patterns of foreign holdings and flows are seemingly consistent with both views. During the global financial crisis, foreign investors' Treasury holdings rose significantly. At the onset of the COVID-19 pandemic, on the other hand, foreign investors sold Treasuries, leading to a new round of debate over whether the special status of Treasuries has waned (He et al., 2022; Weiss, 2022; Vissing-Jorgensen, 2021). As holdings and prices are equilibrium outcomes shaped by the interaction between different investors, however, they do not directly reveal the source of the underlying demand shifts that drive them.

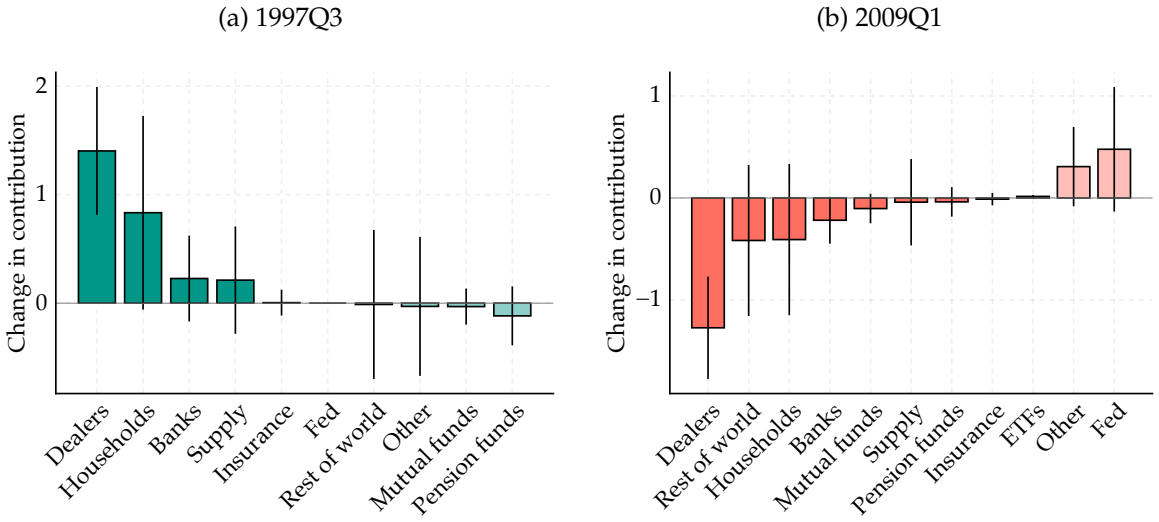
Our approach separates foreign investors' demand response to uncertainty from their response to equilibrium price changes. The estimates support the precautionary saving view. Foreign investors contributed little to the safe-haven property during 1997Q3–2008Q4: their contribution was 0.04, about 2% of the aggregate response of 2.14 (Table 3a). Since 2009, their demand has, if anything, declined as uncertainty rises, making them the largest negative contributor to the safe-haven property in this regime. Compared to domestic investors, the rest of the world's uncertainty loading is 3.84 percentage points below that of domestic private investors in 1997Q3–2008Q4, and 1.60 below since 2009 (Table 3b).

In Appendix Figure B.11, we illustrate the importance of distinguishing observed flows from underlying demand shifts. For both the full sample and the global financial crisis episode, we scale net purchases by foreign investors and by domestic private investors by each group's lagged holdings, then subtract the growth rate of privately held federal debt to account for supply-driven variation in observed flows. On this basis, higher uncertainty is strongly associated with lower foreign flows and higher domestic private flows. Foreign holdings can thus increase while foreign investors absorb propor-

tionately less of the available supply.

Our findings imply that models in which countercyclical demand for U.S. safe assets generates the dollar's exorbitant privilege need another source of countercyclical demand for Treasuries. We next show how domestic participants in the Treasury market such as dealers drive the rise and decline of the safe-haven property.

Figure 5: **Changes in contributions to the safe haven**



Note: This figure plots the change in each sector's contribution to the aggregate total demand loading on uncertainty at 1997Q3 (panel (a)) and 2009Q1 (panel (b)), from the specification of Table 3. Whiskers are 95% confidence intervals, computed as in Table 3.

5.3 Dealers drove both the rise and the decline

Figure 5 plots the change in each sector's contribution at the two regime shifts. Dealers account for the largest changes in the safe-haven property at both shifts. Their demand response to uncertainty rises to 1.31 during 1997Q3–2008Q4, approximately 60% of the aggregate response of 2.14, and is statistically indistinguishable from zero before and after this period. Both the increase in 1997Q3 and the decline in 2009Q1 are statistically significant. The same regime pattern appears in observed dealer flows: dealers' net purchases rise with uncertainty during the middle regime, but show no comparable relationship before or after (Appendix Figure B.12).[24] The household sector, which includes hedge

[24]Primary dealers' cash Treasury positions reported on Form FR 2004 show similar patterns. We estimate a loading on uncertainty at 0.53 over 1998–2008, and obtain estimates close to zero before and after this period. Dealers may hedge cash positions with futures, so a cash-only flow could overstate their response. The historical reports also include futures positions through 2000. Futures changes offset about

funds, exhibits changes in the same direction.

Positive contributions from households and the Federal Reserve sustained the safe-haven property for several years after the decline in dealers' contribution. The household sector contributes 0.40 to the aggregate uncertainty loading after 2009, significant at the 5% level. Still, the rolling estimates in Figure 4 show that its contribution also starts to decline after the early 2010s. After 2009, a one-standard-deviation increase in uncertainty is associated with an estimated increase in Federal Reserve demand equivalent to 0.48% of the Treasury market, although the estimate is imprecise. As the household contribution declines, the aggregate demand loading on uncertainty weakens, with the Federal Reserve accounting for almost all of the remaining response in the final rolling windows.

Because our approach is built to give a bird's-eye view of the whole market, our estimates do not identify why dealers' demand response to uncertainty changed across regimes, but the timing suggests a connection to their balance-sheet capacity. The increase in dealers' demand response to uncertainty in 1997Q3 coincides with the expansion of dealer balance sheets and leveraged relative-value trading in Treasuries, when they had both the capacity and the incentive to absorb Treasuries when risk rose and other investors' demand fell. The decline in dealers' demand response to uncertainty after 2009 coincides with tighter leverage and liquidity regulation, which raised the balance-sheet cost of warehousing Treasuries (Duffie et al., 2023; Cochran et al., 2023; Chaudhary, Fu, & Li, 2026b). This interpretation links the safe-haven property during 1997–2008 to intermediaries' capacity to absorb Treasuries during stress. Subsequent constraints on that capacity may help explain both the decline in the safe-haven property and the reduced private liquidity provision documented in Section 4.[25]

6 Conclusion

This paper asks whether the two properties that underpin the Treasury market's central role, its liquidity and its safe-haven property, are eroding, and which investors are behind the change. Both properties are equilibrium outcomes of investor behavior, so answering the question requires a framework that links the aggregate pricing properties measured by factor regressions to the investors who produce them. To that end, we develop a de-

30% of coupon cash changes, but adding them to cash Treasury positions leaves the estimated response to uncertainty after 1997Q3 nearly unchanged.

[25]Corroborating our finding is Antolín-Díaz (2026), who uses identified flight-to-safety shocks to find that broker-dealers' share of Treasury holdings increases following the tightening, while other sectors do not respond significantly.

mand system in which each sector's Treasury demand responds to prices and to economic factors, and in which market clearing aggregates these responses into the standard price-factor regression. An optimal granular instrumental variable estimator identifies each sector's price elasticity and factor loadings from publicly available holdings data, which lets us follow the market and its investors over four decades.

Applied to 1986 to 2023, the framework yields two conclusions. First, the market's liquidity has declined moderately, despite the significant change in the composition of the investor base. Private investors, in particular foreign investors, have pulled back from providing liquidity since the financial crisis; foreign holdings remain large, but their price elasticity has fallen to near zero, and no private investor has filled the gap. Aggregate liquidity has fallen far less than this retreat implies because the Federal Reserve's post-crisis holdings now account for about a third of the market's aggregate elasticity. Second, the safe-haven property has faded. It emerged only in the late 1990s, was strongest through the crisis, and has since largely faded. Dealers, not foreign investors, drove both the rise and the decline; foreign investors, if anything, sell Treasuries when stress rises, consistent with precautionary saving rather than flight to safety. After dealers stepped back, the household sector and the Federal Reserve's response to stress carried the property for several more years before it faded.

Both conclusions point to the Federal Reserve. Its post-crisis holdings supply about a third of the market's aggregate elasticity, and its purchases in response to stress slowed the decline of the safe-haven property without stopping it. Consequently, properties that private investors once supplied now depend on the size and conduct of the central bank's balance sheet. This dependence matters for balance-sheet normalization. A further reduction in the Fed's holdings removes liquidity that no private investor has shown signs of replacing, and leaves the safe-haven property with a smaller base still. Our estimates characterize these aggregate properties and their sector-level drivers; they do not model the mechanisms, such as regulation, balance-sheet constraints, or reserve management, behind each sector's behavior. Tracing those mechanisms, and asking whether private liquidity provision can be rebuilt, is a natural next step for future work.

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Conflict of Interest

The authors have no conflicts of interest to declare.

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